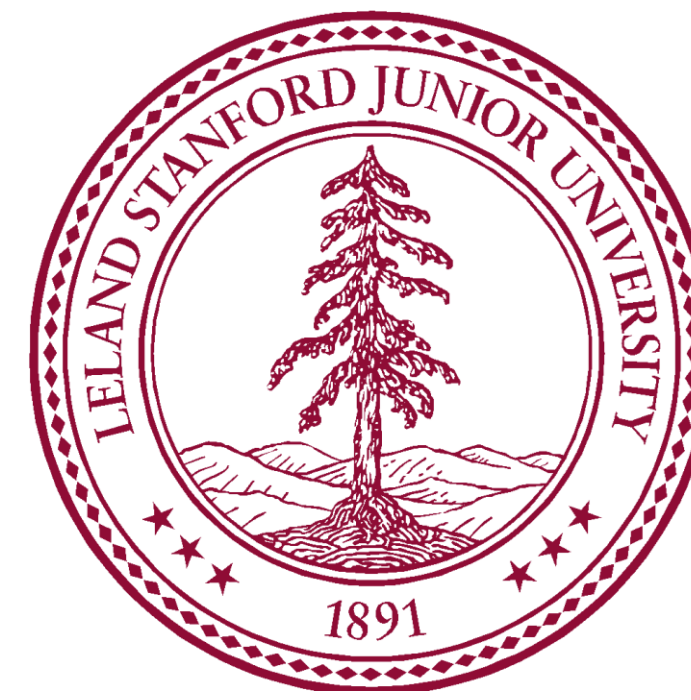


Optimal Sizing of Solar, Wind, and Battery Storage to Compensate for Nuclear Decommissioning in France



Anna-Shérázade Sylla & Matthieu Hautsch - Stanford University

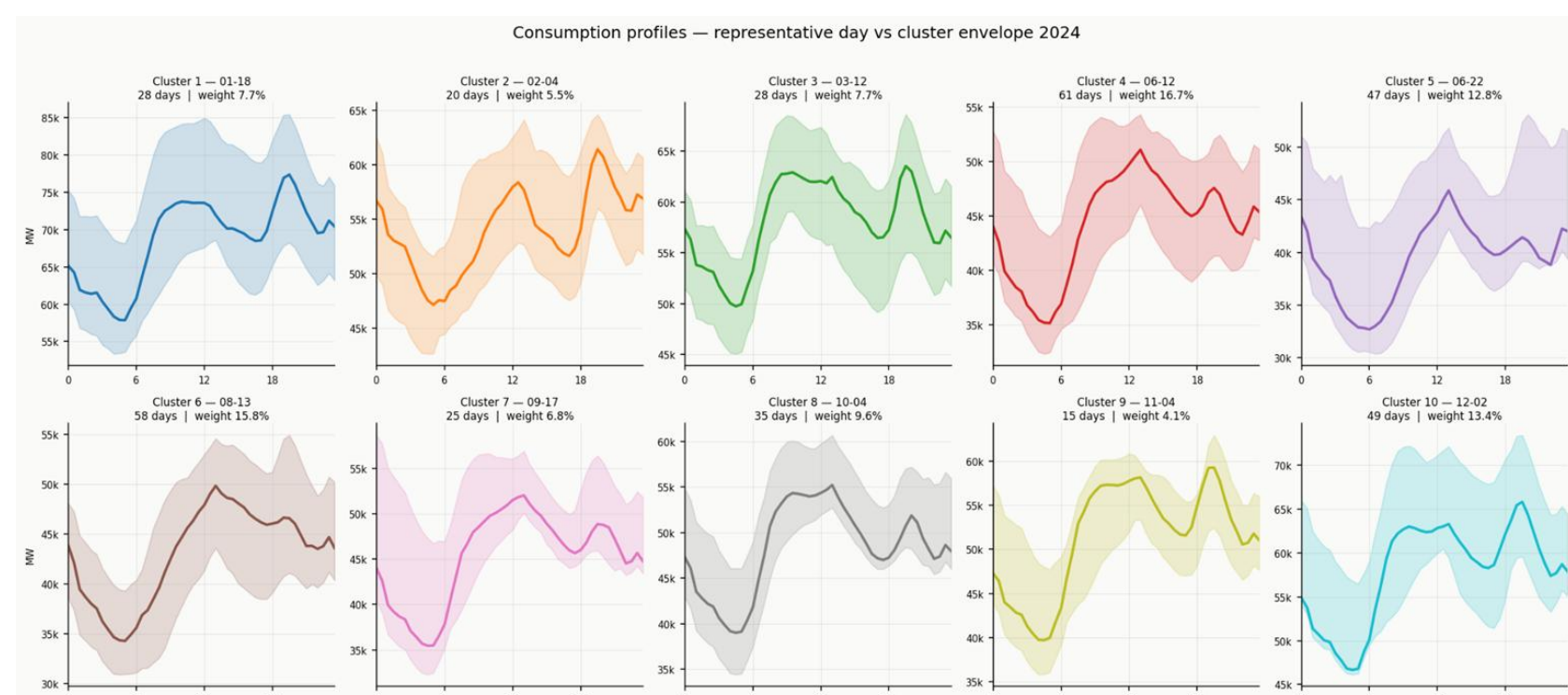
ENERGY 291 Project. Spring 2026

Motivations

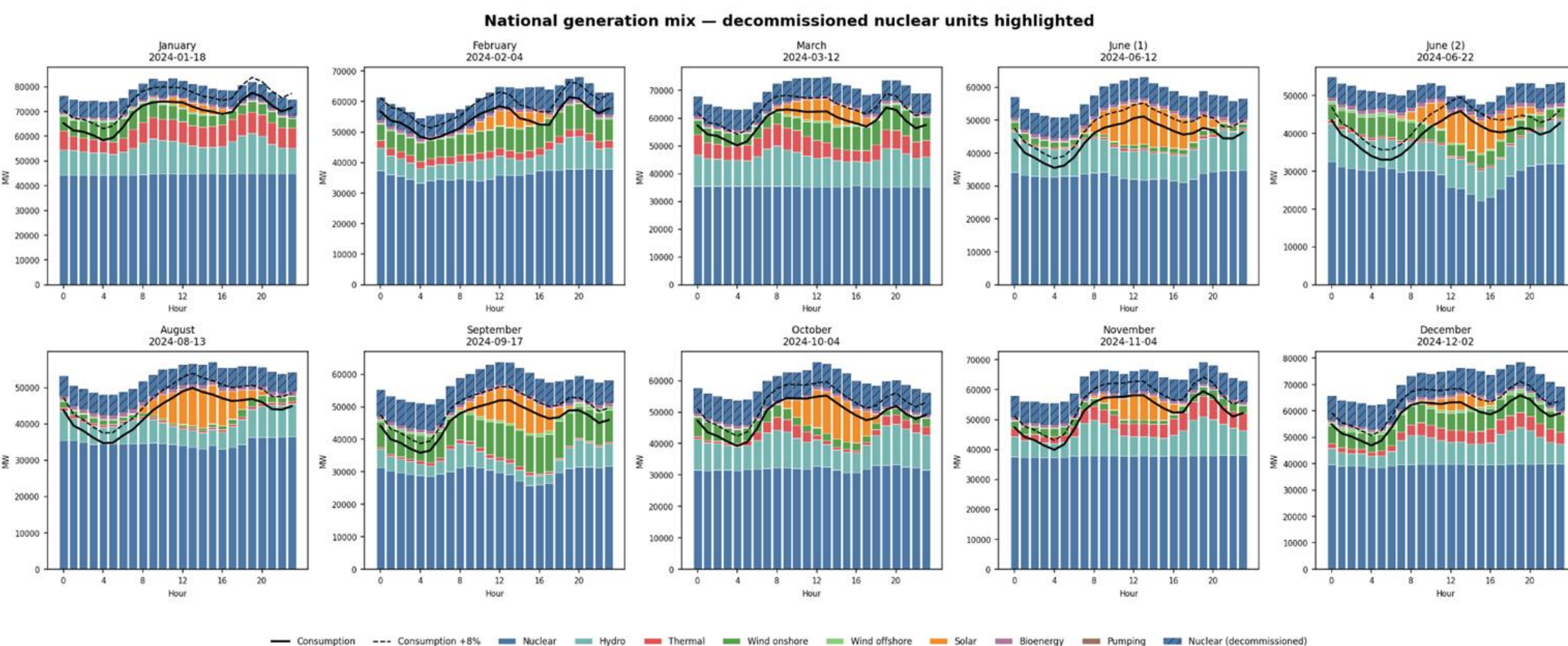
France runs the world's most nuclear-intensive grid (~67% of generation in 2024). Most of its 56-reactor fleet (62.9 GW), built in the 1970s–80s, is nearing its 40-year design life. Under a 50-year lifetime [1], ~10 GW could retire by 2030 (8 million households' demand). We build a cost-minimizing capacity-expansion model for the optimal mix of solar PV, onshore wind, and battery storage to replace it – with no fossil fuels or imports.

Main Assumptions

- System boundary:** Isolated French grid (no imports/exports), 12 regions
- Temporal representation:** 10 representative days via k-medoids clustering of 2024 (50% load, 50% generation mix)
- Demand & generation mix:** Hourly RTE profiles [2,3], scaled +8% for electrification; retiring nuclear removed



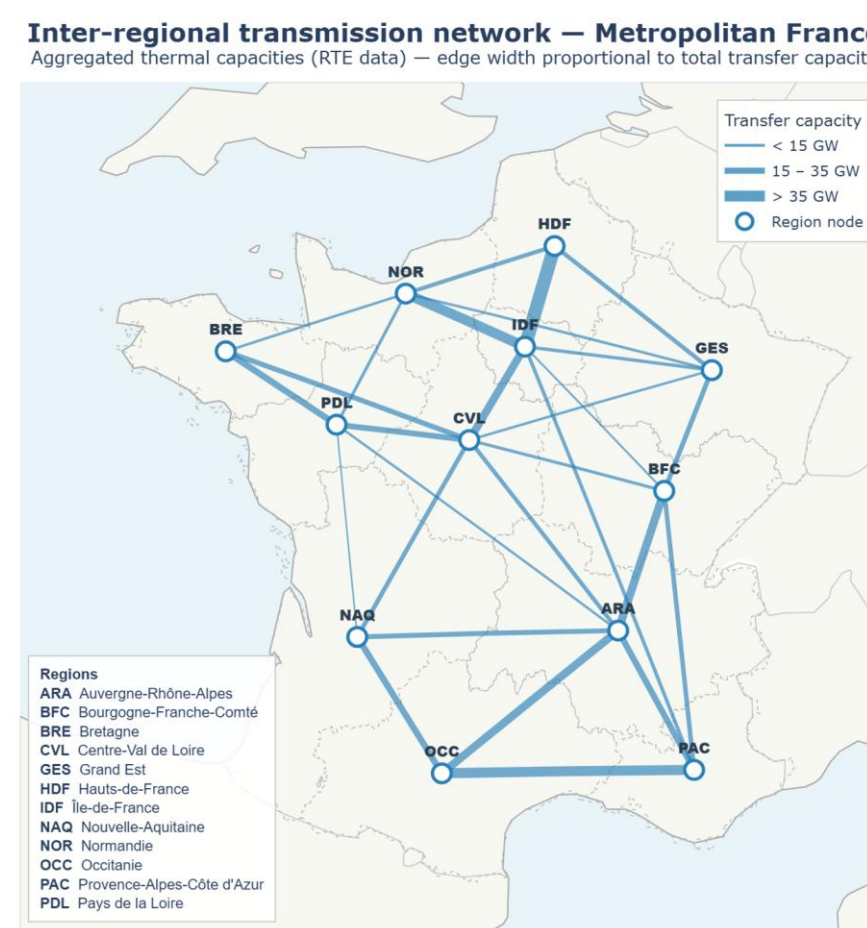
- Renewable generation:** Hourly per-region capacity factors for utility-scale PV and onshore wind [4]



- Storage:** 4-hour Li-ion battery with no efficiency loss; inter-day SOC via Kotzur et al. [5] for multi-week dynamics

- Transmission:** Existing per-region capacity (400 kV: 1,500/1,000 MW; 225 kV: 400/300 MW) [6,7]; new builds overhead only

- Annualized costs:** CAPEX + OPEX at a 6% discount rate [8,9,10]



Technology	Solar PV	Onshore Wind	Battery (4h)	225 kV line	400 kV line
Length of Life (yrs)	25	25	20	40	40
Annualized Cost (€/yr)	84.21 /kW	170.87 /kW	25.53 /kWh	101.83/MW*km	61.10 /MW*km

Model

We use a Mixed Integer Linear Programming model to solve this problem.

Sets & Indices

$r \in \mathcal{R} = \{1, \dots, 12\}$ regions; $d \in \mathcal{D} = \{1, \dots, 10\}$ representative days; $t \in \mathcal{T} = \{1, \dots, 24\}$ hours/day; $i \in \mathcal{I} = \{1, \dots, 366\}$ calendar days; $(r, r') \in \mathcal{P} \subseteq \mathcal{R}^2$ adjacent pairs.

Objective Function

$$\min \sum_r [c^{sol} x_r^{sol} + c^{wnd} x_r^{wnd} + c^{bat} x_r^{bat}] + \sum_{(r,r') \in \mathcal{P}} [y_{r,r'}^{400} c^{400} d_{r,r'} F^{400} + y_{r,r'}^{225} c^{225} d_{r,r'} F^{225}]$$

Decision Variables

- $x_r^{sol}, x_r^{wnd} \geq 0$: new gen. capacity [MW]
- $x_r^{bat} \geq 0$: battery energy [MWh]
- $y_{r,r'}^{400}, y_{r,r'}^{225} \in \mathbb{Z}_{\geq 0}$: new circuits
- $f_{r,r',d,t} \in \mathbb{R}$: power flow [MW]
- $b_{r,d,t}^+, b_{r,d,t}^- \geq 0$: charge/discharge [MW]
- $e_{r,d,t}^{intra}$: intra-period SOC [MWh]
- $\bar{e}_{r,d}^{intra}, e_{r,d}^{+intra}$: SOC min/max
- $e_{r,i}^{inter} \geq 0$: inter-period SOC [MWh]

Energy Balance

$\forall r, d, t$:

$$\tilde{G}_{r,d,t} + CF_{r,d,t}^{sol} x_r^{sol} + CF_{r,d,t}^{wnd} x_r^{wnd} + b_{r,d,t}^- - b_{r,d,t}^+ + \sum_{r':(r',r) \in \mathcal{P}} f_{r',r,d,t} - \sum_{r':(r,r') \in \mathcal{P}} f_{r,r',d,t} \geq D_{r,d,t}$$

Battery — Kotzur et al. (2018)

$\forall r, d, t$:

$$\begin{aligned} e_{r,d,t}^{intra} &= e_{r,d,t-1}^{intra} + b_{r,d,t}^+ - b_{r,d,t}^-, & e_{r,d,0}^{intra} &= 0 \\ e_{r,d}^{intra} &\leq e_{r,d,t}^{intra} \leq e_{r,d}^{+intra}, & b_{r,d,t}^\pm &\leq x_r^{bat} / 4 \\ \sum_{d \in \mathcal{D}} n_d \cdot e_{r,d,t}^{intra} &= 0, & e_{r,i}^{inter} &= e_{r,i-1}^{inter} + e_{r,d(i),T}^{intra} \\ 0 &\leq e_{r,i}^{inter} + e_{r,d(i)}^{+intra} &\leq x_r^{bat}, & e_{r,i}^{inter} + e_{r,d(i)}^{-intra} \geq 0 \end{aligned}$$

Parameters

- $D_{r,d,t}$: demand [MW]
- $\tilde{G}_{r,d,t}$: reduced gen. (excl. retired nuclear) [MW]
- $CF_{r,d,t}^{sol}, CF_{r,d,t}^{wnd}$: capacity factors $\in [0,1]$
- $\tilde{T}_{r,r'}$: existing transmission [MW]
- $d_{r,r'}$: distance $\times 1.25$ (tortuosity factor) [km]
- F^{400}, F^{225} : MW per new circuit
- c^{sol}, c^{wnd} : ann. generation costs [€/MW/yr]
- c^{bat} : ann. battery cost [€/MWh/yr]
- c^{400}, c^{225} : ann. transmission costs [€/MW/km/yr]
- n_d : days in cluster d
- $d(i)$: rep. day of day i .

Transmission

$\forall (r, r'), d, t$:

$$|f_{r,r',d,t}| \leq \tilde{T}_{r,r'} + y_{r,r'}^{400} F^{400} + y_{r,r'}^{225} F^{225}$$

Results & Sensitivity Analysis

Baseline – 8% demand growth

- Wind dominates: 12.6 GW of new capacity
- No solar at all
- Hauts-de-France alone holds 86% of new wind and 100% of battery – highest capacity factors + close to Île-de-France
- PACA is a secondary hub: 1.75 GW of wind
- No new transmission – existing grid suffices

12.6 GW

New wind capacity

86%

Built in Hauts-de-France

0

New transmission lines

8 GW

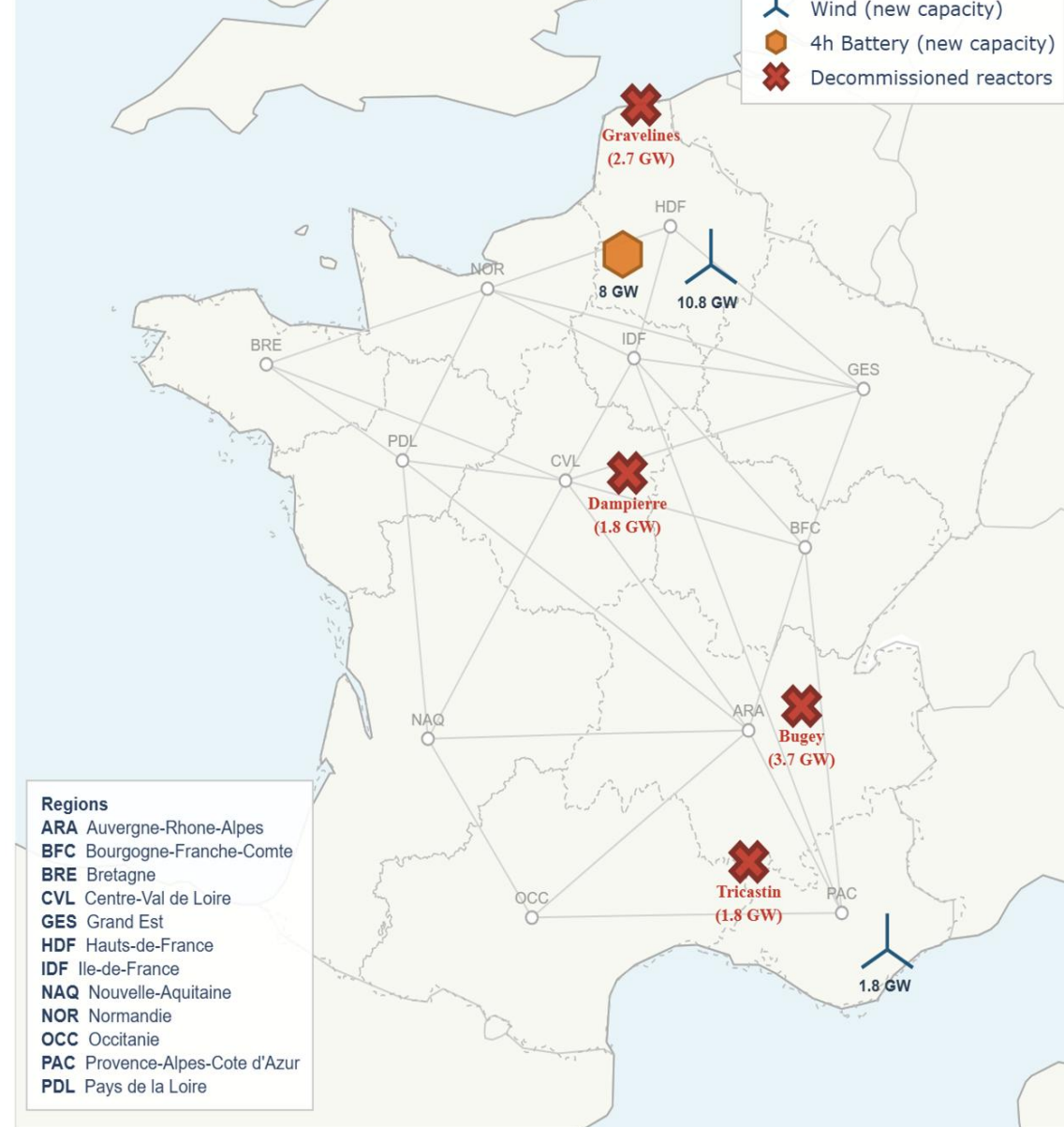
New battery capacity

2.98 B€/yr

Optimal annualized cost

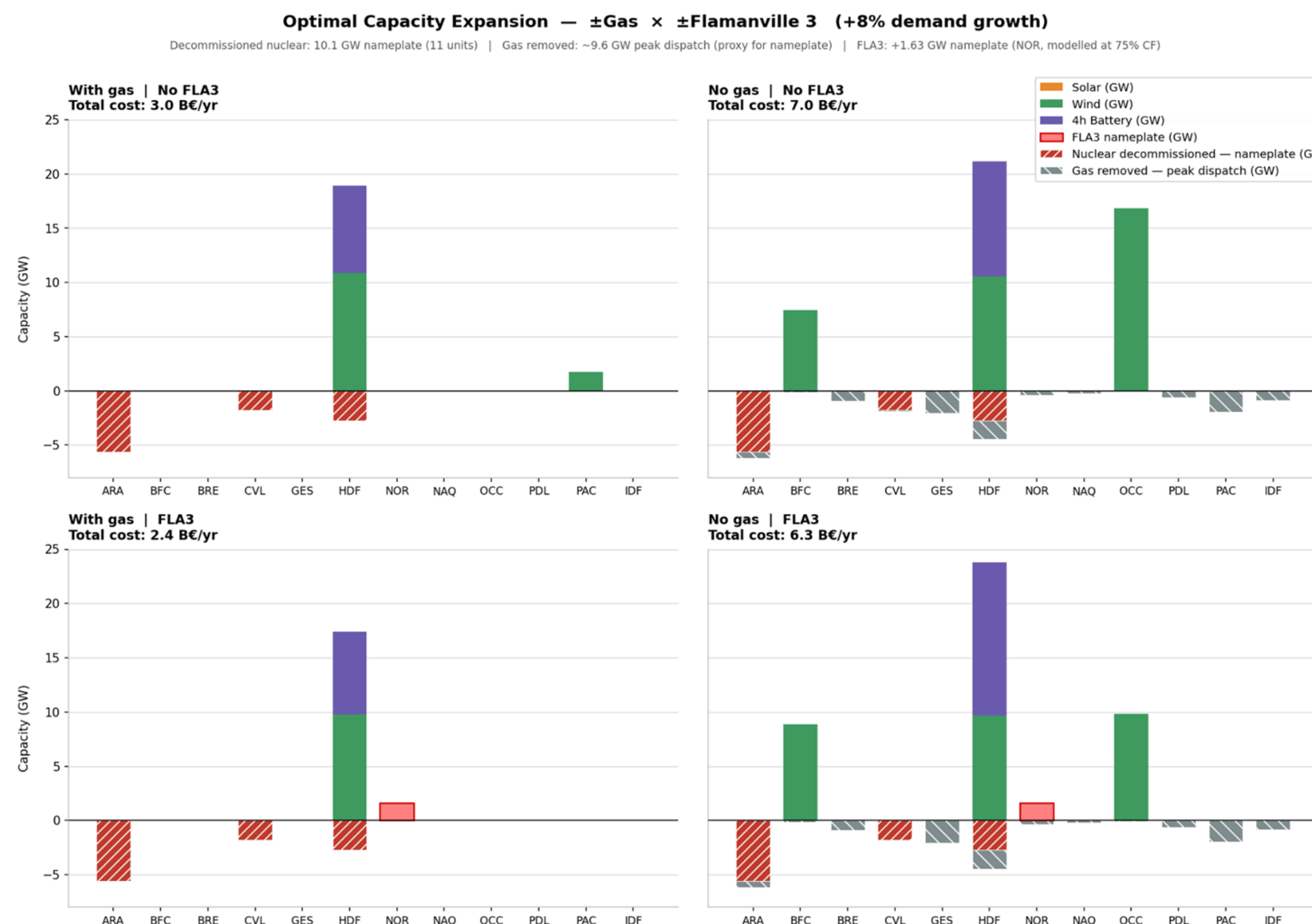
Optimal Investments and Nuclear Decommissioning — Metropolitan France

Scenario: +8% demand growth, constrained transmission | Total decommissioned: ~10.0 GW



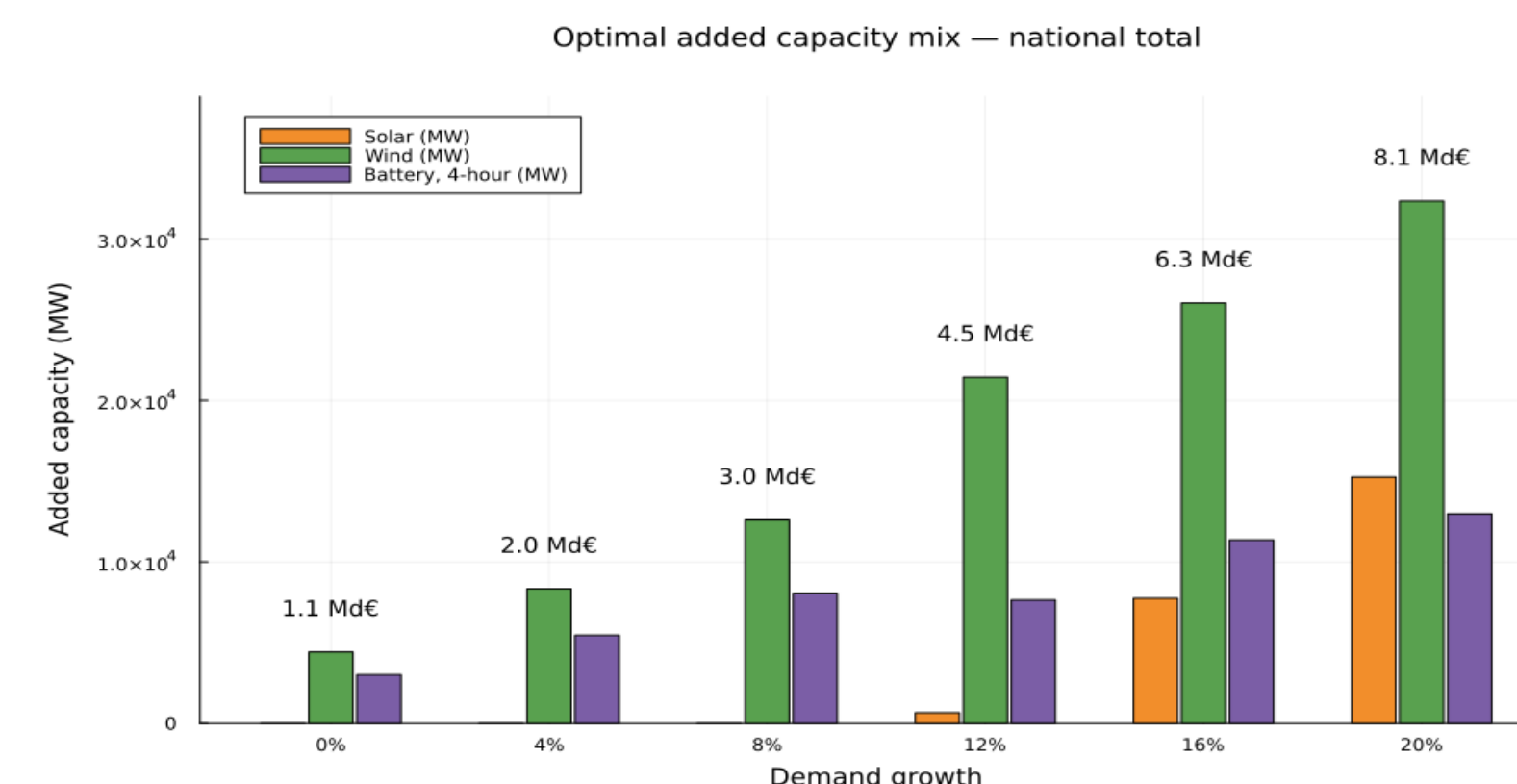
Four scenarios: $\pm$ gas backup, $\pm$ Flamanville 3 (FLA3)

- FLA3 EPR (1.6 GW) connected to the grid in 12/2024
- Removing fossil backup multiplies cost 2.4 $\times$ and needs 2.8 $\times$ more wind
- FLA3 saves 525 M€/yr with gas backup and cuts wind needs 18%
- No-fossil + FLA3 needs more battery than no-fossil alone: flat nuclear baseload misses demand peaks, leaving larger mismatches to buffer
- Without gas, geography shifts – Occitanie and Bourgogne-Franche-Comté enter the optimum



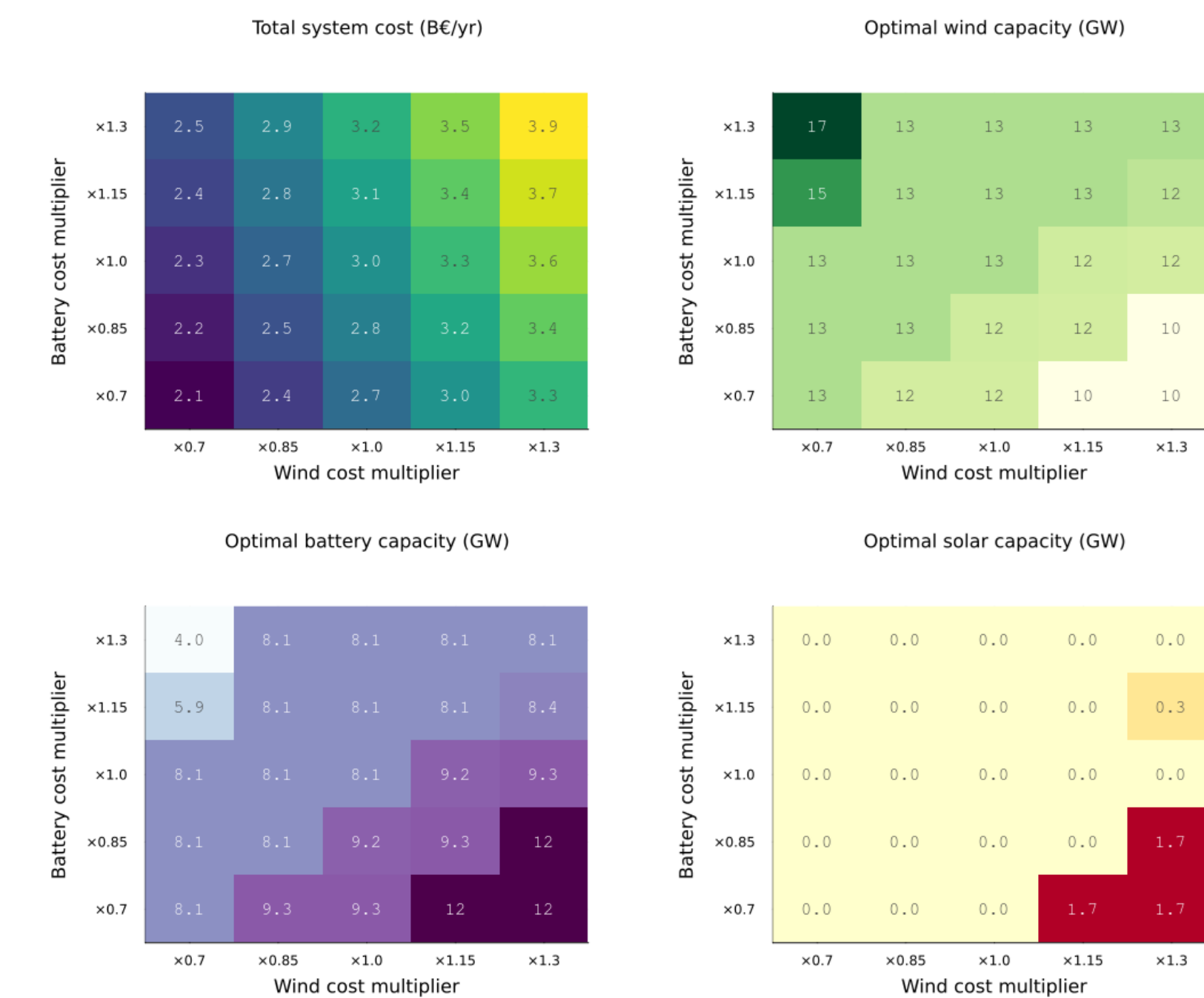
Sensitivity – demand growth (0% $\rightarrow$ 20%)

- Near-linear at low growth: each +1% of demand adds ~270 M€/yr (0–10%)
- Solar appears only above ~12% growth
- Baseline (+8%) sits well inside the wind-only regime – robust to moderate forecast error of the demand growth



Sensitivity – cost heatmaps (wind/battery, $\pm 30\%$)

- Total system cost spans 2.1–3.9 B€/yr across the sweep
- Wind cost dominates: varying it shifts total cost ~54%, battery only ~19%
- Clear wind–battery substitution: cheap battery + costly wind $\rightarrow$ more battery, and vice versa
- Solar enters only at the margin



Conclusion & Future Work

- Wind + battery storage** is the cost-optimal replacement across all scenarios. Solar PV is deployed only above +10% demand, and no new transmission is needed in any scenario – showing the quality of the existing grid.
- Investment is **geographically concentrated**: Hauts-de-France holds 86% of new wind and 100% of battery capacity.
- A fully fossil-free system is feasible but far costlier**: removing gas backup raises annualized costs 2.4 $\times$ and needs 2.8 $\times$ more wind.
- Renewables exhibit lower annualized investment costs than new nuclear under the assumptions considered**: six EPRs at Flamanville cost (79 B€) for the same ~10 GW imply an annualized CAPEX 65% above the renewable baseline (2.98 B€/yr) – though this ignores nuclear's firm, dispatchable capacity.

Future work

- Co-optimize dispatch and investment via a full unit-commitment model
- Add reliability constraints (e.g. Planning Reserve Margin)
- Account for transmission losses and offshore wind
- Implement a more precise transmission network, and increase spatial resolution

References

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